Agentic AI Case Study: Cisco TAC

CASE DETAILS

Company: Cisco Systems

Industry: Networking Technology / Enterprise IT Support

Use Case: AI Agent for Customer Support and Technical Issue Triage

Reference: https://docs.langchain.com/oss/langgraph/case-studies

INFORMATION

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Cisco Systems and the Technical Assistance Center (TAC)

CISCO SYSTEMS

- Builds networking, security, collaboration, and observability products for global enterprises
- Operates across data center, campus, WAN, cloud, and service provider environments
- Customers expect high reliability, strong security, and rapid support response

CISCO TECHNICAL ASSISTANCE CENTER (TAC)

Scale of operations

~1.8M support cases/year

Publicly stated in Cisco/LangChain case materials

- Global engineering support for Cisco products and solutions
- Handles high-volume, multi-domain issues (hardware, software, configuration)
- Coordinates across product teams and knowledge bases (Reasonable inference based on TAC role)

The Problem — Diagnosing Complex Network Issues

- Multi-layer issues span hardware, OS, configuration, and policy domains
- Engineers search across logs, configs, documentation, and live telemetry
- High signal-to-noise: very large log volumes make root-cause isolation difficult
- Triage and escalation can be slow and inconsistent across product teams
- Expertise does not scale linearly with case volume (Reasonable inference based on the described architecture)

SCALE OF OPERATIONS

~1.8M support cases per year

Cited in Cisco/LangChain case materials

DIAGNOSTIC BOTTLENECKS

- Manual parsing of complex, multi-source logs
- Tool fragmentation and cross-team handoffs
- Misconfiguration frequently contributes to incidents (Reasonable inference based on the described architecture)

AI Solution — Multi-Agent Assistant with Supervisor

WHAT CISCO BUILT

- Multi-agent assistants (e.g., Firewall Assistant) to guide diagnosis
- Supervisor/orchestrator decomposes requests and routes to specialized agents
- LangGraph for orchestration of stateful, multi-step workflows
- LangSmith for centralized tracing and observability
- LangChain abstractions for tools/models and standardized agent interfaces
- Retrieval over internal docs and analysis of logs/configs (Reasonable inference based on described architecture)

CORE TOOLS

LangGraph | LangSmith | LangChain

HOW IT WORKS

A single customer query may traverse up to ~6 agentic systems before resolution (reported in Cisco case talks).

Note: Items marked "Reasonable inference" are based on the described architecture

Agent Layer vs Traditional Software

AI / Agent Layer

Reasoning & orchestration

- LLM reasoning engine (planner/dispatcher)
- Supervisor routes to specialized agents
- Multi-step workflows (LangGraph)
- Tool selection logic; RAG over internal knowledge
- Predictive ML models integrated for determinate signals

Traditional Software Components

Deterministic systems

- Telemetry/log pipelines and parsers
- Network monitoring & observability systems
- Internal documentation/KBs; search/vector stores
- Support ticket/case systems; auth/permissions
- LangSmith tracing and audit (observability infra)

INTERACTION FLOW (HIGH-LEVEL)

Engineer query → LLM plans (Supervisor) → Select tools/agents → Call data systems → Interpret & guide → Update ticket & trace

Flow details are a reasonable inference based on the described architecture.

What Is Agentic?

DEFINITION

Agentic AI is an LLM-driven system that autonomously selects actions and tools, reasons across multiple steps with state, and iteratively works toward a goal using available software resources.

In Cisco TAC, the agent plans next actions, invokes tools/agents, and adapts based on results to assist diagnosis.

KEY AGENTIC BEHAVIORS

- **Autonomous tool selection**
  Chooses which tool/agent to call based on current context.
- **Multi-step reasoning**
  Plans and executes sequences of steps rather than one-shot replies.
- **Iterative investigation**
  Analyzes results, refines hypotheses, and repeats until satisfied.
- **Goal-directed behavior**
  Acts to achieve a target outcome (e.g., isolate likely root cause).
- **Memory/context passing**
  Carries forward case context and intermediate findings across steps.

Why this is agentic: the system decides and acts over time using tools and state, not just generating a single text answer.

What Is NOT Agentic?

Deterministic software

Not LLM reasoning

- Telemetry and monitoring systems
- Device logs and log parsers
- Databases, knowledge bases, search/vector stores
- Support ticket and case management systems
- Deployment infrastructure (APIs, services, CI/CD)

Infrastructure used by agents

Deterministic tools

- Observability and tracing (e.g., LangSmith)
- Authentication, authorization, audit logs
- Service gateways and standardized APIs
- Semantic/metadata layers (e.g., Snowflake, Cortex) (Reasonable inference based on the described architecture)

These components are deterministic systems that the agent calls. They do not plan or reason; they provide data or enforce control.

Benefits and Early Results

COVERAGE

60% of ~1.8M support cases

Touched by AI-driven automation

ACCURACY

95%+ accuracy

Renewals risk recommendations (specific use case)

EFFICIENCY

20% time reduction

In renew workflows in under 3 weeks

MATURITY

6+ mo in production

Sustained deployments at scale

QUALITATIVE BENEFITS

- Faster troubleshooting and more consistent triage
- Improved engineer productivity via guided investigation
- Better interoperability through standardized agent interfaces

Note: These qualitative benefits are a reasonable inference based on the described architecture and reported outcomes.

WHY THIS MATTERS

- Agentic AI accelerates diagnosis by orchestrating tools and data
- Standardized observability improves reliability and oversight
- Cross-team reuse of agent workflows reduces duplication

Source for metrics: Cisco TAC case talks (LangChain Interrupt 2025) and LangGraph case studies.

Risks and Limitations

LLM hallucinations

Models may generate confident but incorrect diagnoses or steps.

Mitigation: retrieval constraints, guardrails, and trace-based review.

Incorrect log interpretation

Parsing or summarizing noisy logs can miss critical signals.

Mitigation: structured parsers, schema checks, SME validation.

Data quality dependency

Outdated docs, incomplete telemetry, or sparse tickets degrade outputs.

Mitigation: data freshness SLAs and continuous evals.

Multi-agent complexity

Orchestration across agents can create brittle flows and latency.

Mitigation: standardized protocols and centralized observability.

Human oversight requirement

Engineers must review high-impact recommendations and actions.

Mitigation: approvals, eval dashboards, and escalation paths.

Regulatory/compliance

Access control, auditability, and data residency must be enforced.

Mitigation: role-based access, tracing, and policy checks.

Note: Some mitigations are reasonable inferences based on the described architecture.

Key Takeaways

LESSON 1

Agentic AI = reasoning + tool use + autonomy

The agent plans steps, selects tools, and acts toward a goal.

LESSON 2

Real AI systems are hybrid systems

LLMs work with deterministic software and predictive ML models.

LESSON 3

Scale requires modularity

Use graphs, specialized agents, and standardized interfaces.

LESSON 4

Observability is non-negotiable

Traces, evaluations, and audits are needed for reliability.

LESSON 5

AI augments engineers, not replaces them

Human oversight remains necessary for accuracy and safety.

68% by 2028: Cisco research projects that agentic AI will handle 68% of customer service and support interactions by 2028. (Cisco, 2025)

SPEAKER SCRIPT (5 MINUTES)

1) Context (min 0–1): Introduce Cisco TAC and why network support is complex. State goal: clarify agentic vs traditional components.

2) Problem & Solution (min 1–2): Problems: logs, telemetry, slow triage. Solution: multi-agent assistant using LangGraph workflows, LangSmith tracing.

3) Architecture (min 2–3): Agent layer (LLM reasoning, tool choice, supervisor) vs software layer (telemetry, KBs, tickets). Emphasize interactions.

4) Agentic vs Not (min 3–4): Agentic behaviors: decide tools, iterate, reason. Non-agentic: databases, monitoring, case systems. These are deterministic tools the agent uses.

5) Impact & Risks (min 4–5): Reported results: ~60% cases touched by AI, 95%+ accuracy (renewals), 20% time savings. Risks: hallucinations, data quality; need human oversight. Close with 68% by 2028.

Close: Summarize: Agentic AI orchestrates tools within traditional infrastructure and augments engineers.